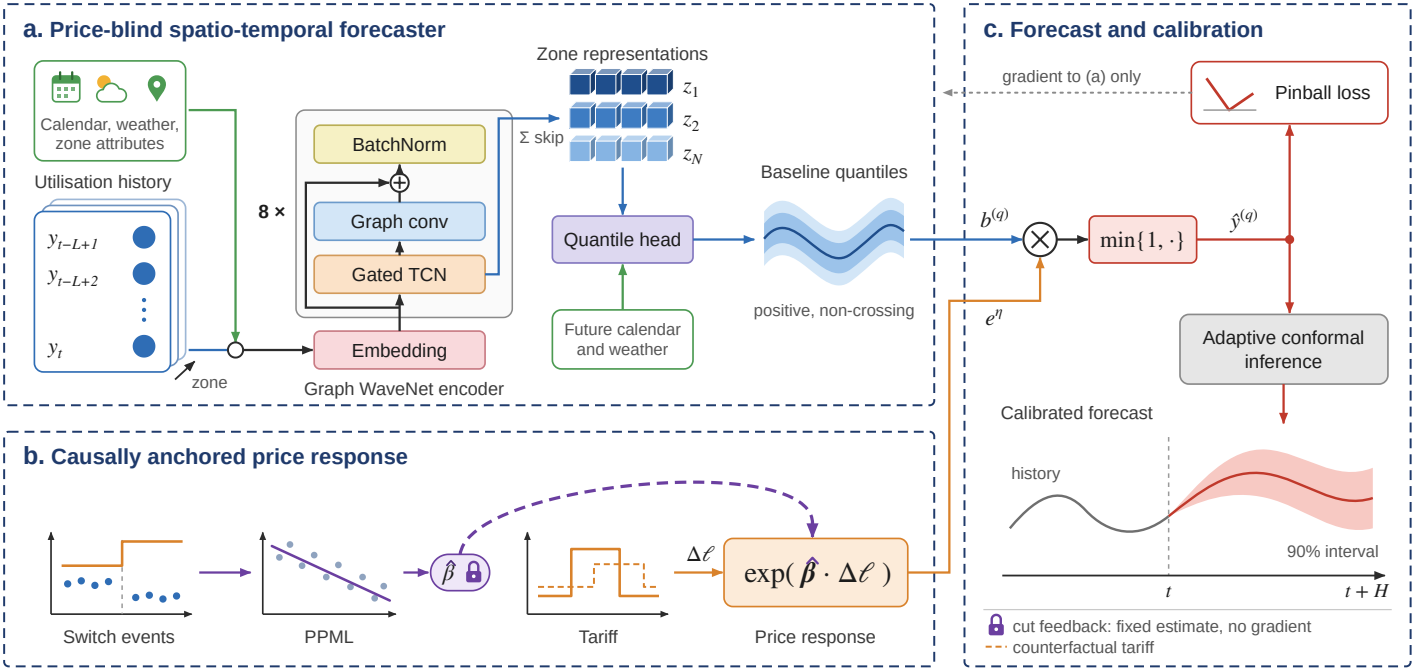

a. Price-blind spatio-temporal forecaster
Calendar, weather, zone attributes
Utilisation history
y_{t-L+1}
y_{t-L+2}
y_t
zone
8 ×
BatchNorm
Graph conv
Gated TCN
Embedding
Graph WaveNet encoder
Zone representations
Σ skip
z_1
z_2
z_N
Quantile head
Future calendar and weather
Baseline quantiles
positive, non-crossing
b. Causally anchored price response
Switch events
PPML
$\hat{\beta}$
Tariff
$\Delta\ell$
$\exp(\hat{\boldsymbol{\beta}} \cdot \Delta\ell)$
Price response
c. Forecast and calibration
gradient to (a) only
Pinball loss
$b^{(q)}$
$\min\{1, \cdot\}$
$\hat{y}^{(q)}$
e^{η}
Adaptive conformal inference
Calibrated forecast
history
90% interval
t
$t + H$
cut feedback: fixed estimate, no gradient
counterfactual tariff